

has sensors measuring and transmitting data about received sunlight and direct current (DC) power generation. These sensors are part of control systems within the plant, with a primary focus on internal sensor functionality.

How does placing all these sensors in a plant impact the monitoring and efficiency?

A solar plant, spread across a vast area, makes fault detection akin to finding a needle in a haystack. Sensors at each stage locate deviations in power generation, offering precise measurements. They generate a loss diagram, detailing power losses from input to final output. Optimising and examining this diagram ensures consistency with engineering predictions, aiding in identifying faults for repair and increasing efficiency.

Do you see the advancements in EV and automotive batteries leading to potential solar power storage space?

Yes, for storage to be truly revolutionised, you will need it at three ends. One, at the plant end, where electricity is generated, to dispatch electricity as and when required. The second storage is needed at the grid end in the substations with a corpus to store excess energy and release it during peak demand periods. Improved battery technology at the consumer end, i.e., the third end, allows consumers to store excess energy and use it as required.

Is there any technology that connects the national grid to battery stations and schedules the power supply?

Yes, there are numerous technologies for this. Scheduling is a smaller aspect of the larger technological innovation happening. Large battery banks for power storage are being built to offer renewable power during peak demand, which may be outside the sunshine/wind hours. The idea is to provide clean power on demand, especially during high-cost periods. Other technologies include finding

ways to store energy more effectively, cheaply, and efficiently to minimise electricity loss. Another critical area is enhancing battery life for multiple cycles, which significantly lowers the overall storage cost.

Do you need any specific technology or a partnership for scheduling?

Partnerships with time-of-day power experts are crucial for accurate predictions of solar power generation, estimating sunlight and grid power needs. In our solar plants, 'digital twins' simulate the energy conversion process via a detailed, virtual model of solar modules. This enables precise forecasting by mirroring the actual asset, identifying areas for improvement. Efficiency gains may involve examining losses from solar modules to transformers. The digital twin learns from past production data, directly impacting revenue for cash-focused investors.

What is the scale of investment you put into all of this technology?

The scale of investment varies. Initially, it primarily involved the upfront capital investment in modules and other components. Nowadays, much of this technology comes built with the modules we purchase. For example, there is something known as string inverters, where each module is tied to an inverter equipped with an inbuilt chip for measurement. Therefore, for us, it becomes a part of the capital investment needed to buy the product. I estimate that it accounts for less than 5% of the system's cost.

Are all the components being acquired domestically or by importing?

While certain tenders are designated for domestic procurement, imports become necessary due to the limited availability of materials. The domestic production capacity for solar modules is much less than the demand, necessitating interna-

tional imports. To discourage reliance on foreign sources when domestic capacity is available, the government has imposed a basic custom duty, making imports 44% more expensive than domestic purchases. Yet, importing becomes the only option once the domestic supply is fully used. This situation is expected to change in the next few years as significant investments are being made in domestic manufacturing, helping bridge the demand-supply gap and allowing for more local production.

Where does India stand in the renewable energy generation ecosystem at a global level?

India, now the fourth-largest in renewable energy generation, has achieved significant progress in the past decade; initially not even ranking in the top 10. The government aims to have 50% of its power from renewable sources by 2030, and with ongoing efforts and observed adoption trends, India may achieve this target even before 2030.

What are the primary challenges in achieving this goal, and how do you propose to face them?

India's renewable energy challenges have evolved. From 2010 to 2015, the focus was on proving solar power viability, now cost-effective and attracting substantial investment. The current challenge is providing 24-hour clean power, addressed by energy storage solutions like traditional batteries or pumped hydro storage. Solar power costs ₹2.5 to ₹3, while storage is at ₹6 to ₹7 per unit, making 24-hour solar slightly costlier than fossil fuels. Despite a 90% reduction in battery pricing over a decade, the remaining 10% is still considered expensive. More efficient and affordable storage (around ₹5 a unit) could provide continuous clean energy at competitive or lower prices than fossil fuels. Minor challenges include payment security and ensuring timely payments for supplied electricity. **EFY**